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It is not because it is offered that it is used: an investigation into firm-level determinants of use intensity of buffering services in science parks

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Abstract Science parks are a type of organizational sponsorship initiative aimed at generating resource-munificent environments through the provision of bridging and buffering services. While extant research has studied the relationship between these services and firm outcomes, it has neglected to provide insights into when tenant firms utilize these services to a greater or lesser extent. Our study evaluates science parks' buffering mechanism and identifies firm-level characteristics affecting tenants' use intensity of buffering services. To do so, we build upon survey data collected from 201 tenant firms located in science parks in three European countries. Taking a resource dependence perspective, our findings indicate that tenants' managerial resources (particularly top management team size and functional experience) and boundary-spanning capacity jointly determine the extent to which firms rely upon the science park's buffering mechanism. This study contributes to the organizational sponsorship, resource dependence, and science park literatures, as well as has important practical implications.

Plain English Summary Which firm-level factors determine the use intensity of science parks' buffering services? We study our research question in a sample of 201 firms located in 35 science parks in Denmark, Spain, and Belgium. Our results show that firms with smaller top management teams or higher levels of functional experience in the top management team use these services to a larger extent. Furthermore, smaller top management teams utilize buffering services to a larger extent if they have greater boundary-spanning capacity. Our study contributes to the ongoing debate on the boundary conditions of organizational sponsorship by providing an understanding of when buffering services are actually used by potential beneficiaries. Accordingly, our study is particularly relevant for science park managers and policymakers.

Keywords Science parks · Buffering · Resource dependence · Organizational sponsorship

JEL Classification M13 · L26 · L52

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1 Introduction

Entrepreneurial firms have the potential to contribute substantially to local and national economic development, but they are typically quite fragile (Audretsch, 1991). Therefore, policymakers have taken initiatives, referred to as "organizational sponsorship" (Flynn, 1993), to mediate the relationship between new firms and their environment. Particularly, organizational sponsorship envisions creating a resource-munificent context, aimed at decreasing new firms' resource dependence and increasing their chances of survival (Amezcua et al., 2013; Autio & Rannikko, 2016). Examples of organizational sponsorship initiatives are government policies in the form of tax incentives or small business loans, initiatives aimed at stimulating equity investments, or the establishment of business incubators, accelerators, and science parks (Amezcua et al., 2013).

The organizational sponsorship literature has identified two focal mechanisms through which these initiatives can affect the development, survival, and performance of entrepreneurial firms. First, through the bridging mechanism, sponsorship connects new firms to their environment, thus acting as brokers and facilitating network building (Amezcua et al., 2013; Sapsed et al., 2007). Bridging allows firms to actively interact with their environment, providing access to external resources in order to improve their competitive positions (Flynn, 1993). Second, the buffering mechanism involves the provision of necessary resources internally, thus boosting sponsored firms' resource pool and reducing their dependence upon the external environment (Amezcua et al., 2013; Hall, 1982). Through this mechanism, nascent firms are sheltered from potential threats and uncertainty in their environment (Lynn, 2005; Thompson, 1967), and this protection allows them to focus on internal development (Jourdan & Kivleniece, 2017).

We identify two important gaps in the current state of knowledge on organizational sponsorship that led to our research question. First, the buffering mechanism, despite being at the center of organizational sponsorship, remains largely understudied. While it is widely acknowledged in the literature that buffering services are offered across organizational sponsorship initiatives (e.g., Amezcua et al., 2013; Breivik-Meyer et al., 2020; Westhead, 2021), we still lack insights into when (i.e., under which conditions) such sheltering support is most sought after and valued by nascent firms. Closer scrutiny inside the buffering mechanism is warranted as it may have important implications for sponsored new firms. Buffering support implies access to physical infrastructure and professional services (e.g., affordable office space, shared meeting rooms, administrative support), which could lower the costs for new firms and ease the challenge of efficiently organizing their core operations (Woolley & MacGregor, 2021). By buffering the negative effects of internal resource scarcity and dependence on external resource providers (Autio & Rannikko, 2016; Pfeffer & Salancik, 1978; Wiklund et al., 2010), sponsorship can enhance firm performance and survival (Amezcua et al., 2013, 2020). Furthermore, resource buffering through organizational sponsorship may create a competitive advantage over other industry actors or competitors that did not receive it (Jourdan & Kivleniece, 2017; Lazzarini, 2015). Therefore, our study aims to unravel antecedents of the buffering mechanism of organizational sponsorship.

Second, extant research into organizational sponsorship has mainly shed light on the relationship between organizational sponsorship and firm outcomes, alongside the contingencies of the relationship (Amezcua et al., 2013, 2020). Prior studies have largely considered whether (or not) organizational sponsorship is available to firms, for example, whether (or not) incubators offer specific services (e.g., Amezcua et al., 2013) or whether (or not) firms are located in an incubator (e.g., Amezcua et al., 2020). By doing so, the literature has taken the implicit assumption that, if organizational sponsorship is available, firms will use and benefit from such initiatives. However, in line with recent allusions by Breivik-Meyer et al. (2020) and Van Weele et al. (2017), we argue it is critical to investigate use intensity, or the extent to which the mechanisms are used by firms, to evaluate the efficacy of organizational sponsorship initiatives. Following resource dependence theory, we posit that firm-level characteristics may explain heterogeneity in the use intensity of organizational sponsorship, particularly the buffering mechanism provided by these initiatives.

In targeting our research objectives, we focus on one exemplary form of organizational sponsorship, namely science parks ("SPs" hereafter). SPs are an appropriate empirical setting for our study for at least two reasons. First, SPs constitute a mature and sizable context to study organizational sponsorship. Since their origin in the 1950s, SPs have proliferated considerably worldwide (Link & Link, 2003; Vásquez-Urriago, et al., 2016b). The global proliferation is also evident from the existence of an international industry association (i.e., IASP) with over 350 member organizations in 78 countries. Second, practitioners and scholars have underscored buffering as a central aim for SPs globally. For instance, the IASP definition emphasizes that "science parks provide value-added services together with high-quality space and facilities" (IASP, 2020).[1] Similarly, Phan et al. (2005) defined science parks are property-based initiatives with identifiable administrative centers, established to provide (access to) resources. Indeed, the provision of facilities (e.g., office space, labs, meeting or exhibition rooms, and restaurants) and services (e.g., administrative, managerial, or technological support) is at the core of science park offerings (Ng et al., 2019). The low-cost or shared-use thereof allows tenant firms to minimize costs (Dettwiler et al., 2006) and buffers their core operations by reducing dependence on the external environment for resources (e.g., Amezcua et al., 2013). The widespread engagement in SP establishment as an organizational sponsorship initiative and the common offering of buffering services to tenant firms make SPs a particularly relevant research context for our research question: "*Which firm-level conditions affect tenants' use intensity of the SP's buffering services?*".

To address this research question, we use a combination of survey and archival data from 201 tenants located on 35 SPs in three European countries. Structural equation modeling provides insights into the firm-level conditions affecting the use intensity of the SP's buffering services by tenants. Specifically, we find that tenants with smaller top management teams and with higher levels of functional experience in the top management team rely more extensively on the buffering mechanism. At the same time, the tenant firms' boundary-spanning capacity, or their ability to reach out into the environment to access resources and knowledge, moderates the relationship between team size and the use of the buffering mechanism.

Our study contributes to organizational sponsorship theory, resource dependence theory, and the science park literature. First, our study contributes to the organizational sponsorship theory (Flynn, 1993), which has pointed to the existence of important (environmental) contingencies in the extent to which organizational sponsorship affects firm performance (Amezcua et al., 2020). We advance the ongoing debate on the boundary conditions of organizational sponsorship by pointing to the relevance of firm-level contingencies, specifically with regard to TMT characteristics. Second, our study enriches resource dependence theory (Pfeffer & Salancik, 1978) by indicating under which conditions firms are more or less likely to reduce or overcome their resource dependence by calling upon organizational sponsorship. Finally, our study also contributes to the science park literature by providing insights into the circumstances under which science parks' buffering services are used to a greater or lesser extent. This may provide a more nuanced understanding of performance-enhancing effects (not) resulting from SP localization. Indeed, while some studies find SPs to contribute to tenants, others refute this assertion (e.g., Colombo & Delmastro, 2002; Radosevic & Myrzakhmet, 2009; Squicciarini, 2008, 2009; Vásquez-Urriago et al., 2014; Westhead, 1997). Additionally, this paper provides practical implications for SP managers, (prospective) tenants, and public policy.

2 Theoretical framework

2.1 The relationship between firm managerial resources and the use intensity of SP buffering services: a resource dependence perspective

Resource dependence theory characterizes firms as open systems, dependent on their external environment for the supply of critical resources (Pfeffer & Salancik, 1978). Resource attraction is not only vital for firm survival, but also for growth, successful innovation, and competition in today's global markets (Aldrich & Fiol, 1994; De Massis et al., 2018). Therefore, the managers' ability to manage the dependence of the firm on the external environment is crucial (Pfeffer & Salancik, 1978). At the same time,

[1] Besides the buffering mechanism, the IASP definition also includes the bridging mechanism as a core objective, stating that science parks aim 'to stimulate and manage the flow of knowledge and technology' as well as 'to facilitate communication between companies' (IASP, 2020).

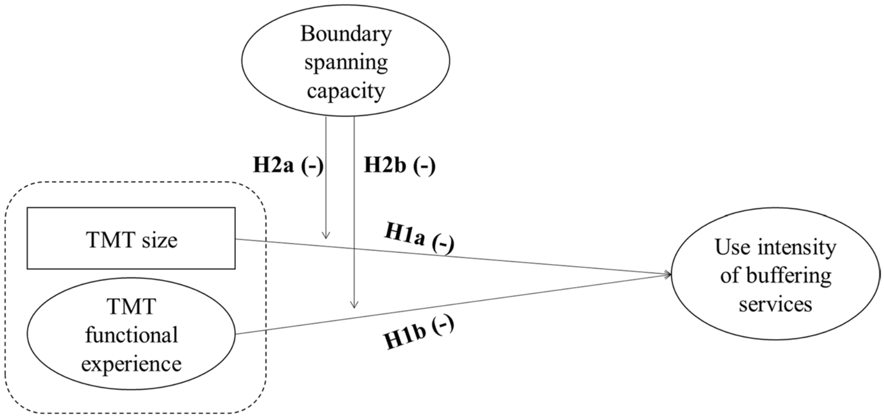

Fig. 1 Conceptual framework

the external environment is characterized as a locus of potential liabilities and threats (Amezcua et al., 2013; Lynn, 2005; Shepherd et al., 2000).

In response, SPs typically provide their tenants with (access to) resources by offering them a broad range of services and facilities within the supporting SP environment (Salvador, 2011). Particularly, the buffering services offered by SPs can be classified along four categories (Rowe, 2014). First, property-related services include the provision of infrastructure and shared services such as conference and meeting rooms, restaurant, and cafeteria (Guy, 1996; Rowe, 2014; Sofouli & Vonortas, 2007). The use of property-related services lowers overhead costs due to economies of scale (Benneworth & Ratinho, 2014; Siegel et al., 2003b; Westhead & Batstone, 1999) and minimizes challenges and burdens associated with the practical side of entrepreneurship (McAdam & McAdam, 2008). Second, business support services are often offered on park (Kihlgren, 2003; Klofsten & Jones-Evans, 1996; Löfsten & Lindelöf, 2003). These services encompass the provision of management, financial, marketing, and legal advice or consulting, and are expected to leverage the tenants' management and business competences (Klofsten & Jones-Evans, 1996). Third, SPs may offer innovation services such as the provision of access to specialized laboratories and scientific equipment, advice on R&D, and technology transfer services. These innovation services are for instance believed to increase the tenants' knowledge base (Salvador, 2011) and lower overhead costs. Finally, networking services include the organization of networking occasions and events that provide access to potential customers, suppliers, funding sources, and partners within or beyond the SP. These events are set up to increase the number of tenants' network contacts and facilitate networking and knowledge sharing (Koçak & Can, 2014).

By offering these services, SPs may buffer the tenants' resource dependence upon the wider external environment outside the SP (Amezcua et al., 2013), mitigate the tenants' resource scarcity and shelter them from environmental threats and uncertainty (Pfeffer & Salancik, 1978). However, we contend that the extent to which this occurs is likely highly contingent on the extent to which tenants use this buffering mechanism offered by SPs (Breivik-Meyer et al., 2020; Van Weele et al., 2017). In what follows, drawing on resource dependence theory, we develop hypotheses on which firm-level conditions may affect the firm's use intensity of buffering services. Figure 1 summarizes our conceptual model.

According to resource dependence theory, the level of dependence of a firm on the external environment is a direct function of its managerial resources and capabilities (Pfeffer & Salancik, 1978). Top management teams (TMTs) are strategic decision-makers within firms (Amason & Sapienza, 1997; Finkelstein & Hambrick, 1996), and play a vital role in establishing and maintaining links between the firm and its environment (Pfeffer & Salancik, 1978). TMTs will attempt to access, acquire, or develop resources that are strategic or scarce, and internally lacking (Huse, 2005), and to minimize dependence in doing so. Thus, managerial resources are likely to affect the need for the buffering mechanism, or the extent to which tenant firms use this type of services on the SP. Particularly, an abundance of managerial resources will lower the firm's resource dependence, and the need for reaching out to the environment (amongst others the SP) for resources, while a scarcity of managerial resources will initiate the opposite mechanism. Two key indicators of the collective managerial resources available to a firm are TMT size and TMT functional experience (Menz, 2012). We consider these two facets of managerial resources also particularly relevant for understanding the firm's resource dependence, and its use intensity of buffering services.

First, TMT size is a frequently used indicator of a firm's level of managerial resources (e.g., Knockaert et al., 2015): larger TMTs have larger levels of managerial resources at their disposal (Ucbasaran et al., 2003). This is because each TMT member enhances the firm's pool of resources by adding skills, knowledge, and network contacts (Shane & Stuart, 2002). Hambrick and D'Aveni (1992, p.1449) state it as follows: "At a basic level, the resources available on a team result from how many people are on it". Also, having more TMT members typically adds to the diversity in skills and knowledge of the team (Bantel & Jackson, 1989; Wiersema & Bantel, 1992). Therefore, tenants with larger TMTs possess more managerial resources, making them less dependent on their external environment for the attraction of resources. By contrast, tenants with smaller TMTs generally possess lower levels of managerial resources and will be more dependent on their environment to acquire and generate resources. However, since the external environment harbors many threats (Shepherd et al., 2000), they will make more use of services offered on the SP, and thus rely more on the buffering mechanism of SPs. Accordingly, we offer the following hypothesis:

Hypothesis 1a. There is a negative relation between TMT size and the use intensity of buffering services.

Second, management teams differ significantly from one another in their levels of knowledge and experience. TMT functional experience refers to the extent to which the firms' TMT members have accumulated experience in different functional domains (Daellenbach et al., 1999). TMT functional experience represents how managers interpret and make sense of the firm's external environment (Day & Lord, 1992). Particularly, TMTs with high levels of functional experience possess diverse and deep knowledge bases and have a broader pool of resources at hand from which to draw in making decisions and taking action (Bunderson & Sutcliffe, 2002; Kor, 2003). Therefore, tenant firms with greater TMT functional experience are less likely to rely on the buffering mechanism of SPs to attract necessary resources. By contrast, tenants that possess lower levels of TMT functional experience have fewer managerial resources at their disposal and will be more dependent on their environment to attract resources. As such, these tenants will more frequently rely upon the SP buffering mechanism and thus use more SP services. We offer the following hypothesis:

Hypothesis 1b. There is a negative relation between TMT functional experience and the use intensity of buffering services.

2.2 The moderating role of boundary-spanning capacity

Thus far, following a resource dependence logic, we expect firms with small TMTs and limited functional experience to use the SP buffering function to a greater extent. Yet, the seminal work by Thompson (1967), which informed resource dependence theory and where the notion of organizational buffering from external pressures and uncertainty originated, indicated the importance of accompanying boundary-spanning functions. Subsequent literature (e.g., Faraj & Yan, 2009; Lynn, 2005; Yan & Louis, 1999) put forward buffering and boundary spanning as distinct forms of boundary work, or the activities a firm engages in to manage firm-environment boundaries, and which are necessary for firm survival (Link & Scott, 2003). We argue that engagement in one type of boundary work (i.e., boundary-spanning) will coincide with the extent to which managers decide to rely on other types of boundary work (i.e., buffering). Whereas buffering represents an action to protect core operations, boundary-spanning entails reaching out to secure needed resources (Faraj & Yan, 2009).

Boundary-spanning capacity is defined as a firm's ability to access resources and knowledge from outside the boundaries of the organization (Rosenkopf & Nerkar, 2001). Particularly, the possession of high boundary-spanning capacity enables firms to undertake boundary-spanning activities, which are actions to reach out into the external environment to obtain critical resources and information, to influence external actors, and to build broader networks (Faraj & Yan, 2009). Furthermore, firms with high boundary-spanning capacity will encourage their employees to solicit critical information and resources from outside the boundaries of the firm and build relational networks (Faraj & Yan, 2009).

We expect tenants' boundary-spanning capacity to function as a moderator in the relationship between tenants' managerial resources and use intensity of SP buffering services. Particularly, we expect tenants to rely more on the buffering mechanism, and thus to use these SP services more extensively, if they have low levels of managerial resources and when they also possess high levels of boundary-spanning capacity. This is because, while tenants with low managerial resources perceive a need for SP services, their boundary-spanning capacity also enables them to do so. This is because using the SP services requires them to cross the borders of their own organization. As such, boundary-spanning capacity will moderate the negative relationship between the tenants' managerial resources and use of SP services. We therefore propose the following hypotheses

Hypothesis 2a. *Boundary-spanning capacity will moderate the negative relation between TMT size and the use intensity of buffering services. Specifically, smaller teams will use buffering services to a larger extent if they have higher levels of boundary-spanning capacity.*

Hypothesis 2b. *Boundary-spanning capacity will moderate the negative relation between TMT functional experience and the use intensity of buffering services. Specifically, TMTs with less functional experience will use buffering services to a larger extent if they have higher levels of boundary-spanning capacity.*

3 Methodology

3.1 Data and sample

We constructed a database containing information on SPs and their tenants in three European countries. The three countries were selected based on their categorization in the Innovation Union Scoreboard (2017)[2] and their geographical location: Denmark ('innovation leader', Northern Europe), Belgium ('strong innovator', Central Europe), and Spain ('moderate innovator', Southern Europe). Subsequently, within each country, we randomly selected one or two NUTS-1 regions: Denmark (DK0),[3] Flemish Region (BE2), Community of Madrid (ES3), and East Spain (ES5). Through web searches and information from overarching institutions and associations, a total of 42 SPs were identified within these four regions. We contacted these SPs and conducted on-site interviews with their general or managing director. Subsequently, using information found on the SP websites and the Orbis Europe database,[4] we identified 2673 tenants on these SPs. For 2255 tenants, an email address was found on the firms' websites. In the spring of 2018, we sent out an online survey to these tenants, specifically addressed to the manager in charge of the firm on the SP. After an email reminder after 2 weeks and a phone call another 2 weeks later, 368 survey responses were returned (or 16%). We eliminated the responses of 42 tenants that indicated to be a public research institution or public organization. Another 20 responses were eliminated as the firms indicated not to be physically located on the SP. Finally, 105 responses were eliminated due to missing responses to the core variables in this study. As such, our final sample consists of 201 responses from tenants in 35 different SPs. To obtain additional information on the tenants, the survey data were merged with archival data retrieved from the Orbis Europe database.

As recommended by Podsakoff et al. (2003), we implemented procedural remedies in the study's design to minimize common method bias. Particularly, we obtained measures from different sources and methodologically separated the measures in the survey by using different response formats. For instance, we complemented latent variables

3 Denmark has only one NUTS-1 region, comprising the whole country.

2 The European Innovation Scoreboard provides a comparative analysis of innovation performance in the European countries. More information: https://ec.europa.eu/growth/industry/innovation/facts-figures/scoreboards_en.

4 The Orbis Europe database is managed by Bureau Van Dijk and contains high-quality general and accounting data on European firms (Vanacker et al., 2017).

Table 1 Tests for non-response bias

Latent variable	Groups	Mean (s.d.)	p value
TMT functional experience	After round 1	5.23 (1.24)	0.051
	After round 2	4.86 (1.38)	
Use intensity of buffering services	After round 1	3.29 (1.23)	0.725
	After round 2	3.23 (1.43)	
Boundary-spanning capacity	After round 1	5.02 (1.26)	0.087
	After round 2	4.69 (1.42)	

Group after round 1, $n=79$

Group after round 2, $n=122$

(multiple-item variables) with observed variables and used different scale formats and anchor points (Podsakoff et al., 2003). Our survey was also pretested, and social desirability was minimized by ensuring anonymity for all respondents.

Further, we examined potential non-response bias in two ways. First, we employed Armstrong and Overton (1977)'s procedure and compared the responses completed after the first email with those completed after the second email. Particularly, we tested mean differences in all multiple-item variables used in this study (Zeng et al., 2010). As Table 1 shows, the results of the t tests show no significant differences ($p>0.05$). Second, we compared general characteristics of the tenants that responded to our survey to those of the tenants that did not respond. Specifically, we compared tenant age, size (in terms of log of total assets), and sector and found no significant differences ($p>0.05$). We can conclude that non-response bias is unlikely a concern.

3.2 Measures

Table 2 provides the means, standard deviations, and correlations for the variables. Appendix Table 5 presents the items of our latent variables.

3.2.1 Dependent variable

Use intensity of buffering services Use intensity reflects the frequency of the usage of SP buffering services at the firm level. Following Rowe (2014), we classify SP buffering services into four categories, namely property-related services, business support services, innovation services, and networking services. For each of these services, we asked respondents to indicate the extent to which they have used them since their establishment on the SP, hereby using a 7-point Likert scale (1 = to a very little extent, 7 = to a very large extent). We used these survey inputs to construct a summated scale, by averaging the scores received for the four services. The Cronbach alpha of this construct is 0.76. The average value of the use of SP services is 3.25 (s.d. 1.35).

3.2.2 Independent variables

TMT size Size was measured as the number of people in tenants' TMT. Following Amason (1996), we specified in the survey that TMT refers to the group of managers or executives who take strategic decisions and who are responsible for the management of the company, hereby excluding persons only sitting on the board of directors/advisors. The average TMT size of the tenants in our sample is 2.90 (s.d. 2.21).

TMT functional experience The six items to assess the functional experience of tenants' TMT were based upon Cantner et al. (2010). Specifically, we asked respondents to indicate the extent to which the TMT members have experience in six functional areas: 'management', 'marketing, sales and promotion', 'accounting, controlling and financing', 'engineering and R&D', 'production', and 'personnel' (1 = to a low degree, 7 = to a high degree). However, following the results of our principal components analysis (PCA) and confirmatory factor analysis (CFA), two items (FE4 'engineering and R&D' and FE5 'production') did not meet the minimal thresholds for factor loadings (i.e., in PCA and CFA, factor loadings should be 0.50 or higher (Hair et al., 2014)). These two items were therefore removed from our

Table 2 Mean, standard deviations, and correlations

Variables	Mean	s.d	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Firm age	10.37	11.03	1														
2. Firm size	14.09	41.39	0.25*	1													
3. Sector IT	0.39	0.49	−0.09	−0.07	1												
4. Sector life sciences and chemical	0.24	0.43	−0.03	0.16*	−0.46*	1											
5. Sector material	0.11	0.32	0.20*	−0.04	−0.29*	−0.20	1										
6. Sector other	0.25	0.43	−0.01	−0.05	−0.46*	−0.33*	−0.21*	1									
7. Belgium	0.17	0.38	0.27*	0.31*	−0.06	−0.01	0.09	0.02	1								
8. Denmark	0.48	0.50	−0.12	−0.18*	0.06	−0.06	−0.10	0.07	−0.44*	1							
9. Spain	0.35	0.48	−0.09	−0.06	−0.01	0.07	0.03	−0.08	−0.33*	−0.71*	1						
10. SP age	25.99	15.21	0.14*	0.07	−0.03	−0.14*	0.07	0.12	0.08	0.49*	−0.58*	1					
11. SP size	88.97	39.19	0.08	0.02	−0.05	−0.09	0.01	0.14	−0.30*	0.26*	−0.04	0.36*	1				
12. TMT size	2.90	2.21	0.25*	0.32*	−0.08	0.08	−0.01	0.02	0.14*	−0.24*	0.14	0.01	−0.03	1			
13. TMT functional experience	5.00	1.34	0.20*	0.13	−0.08	−0.01	0.00	0.10	0.03	−0.05	0.03	−0.02	0.00	0.23*	1		
14. Use intensity of buffering services	3.25	1.35	−0.24*	−0.14	−0.03	−0.06	0.03	0.06	−0.26*	0.11	0.09	−0.04	0.14	−0.16*	0.12	1	
15. Boundary-spanning capacity	4.82	1.37	−0.07	0.02	−0.16*	−0.02	0.09	0.01	0.03	0.20*	−0.24*	0.08	−0.16*	−0.010	0.15*	0.18*	0.16*

Significance level: $*p < 0.05$ or lower; $n = 201$; correlations between dummy variables should be interpreted with care analyses. The Cronbach alpha of the final construct consisting of four items is 0.78. We obtained this construct by averaging the replies received for these four items. The average value of the functional experience of the tenants' TMT is 5.00 (s.d. 1.34).

3.2.3 *Moderator variable*

Boundary-spanning capacity We adopted the validated scale developed by Faraj and Yan (2009), which consists of four items we slightly adjusted to fit the SP context. We averaged the scores received for these four items for obtaining the moderator. The Cronbach alpha of this construct is 0.86. The average value for tenant boundary-spanning capacity is 4.82 (s.d. 1.37).

3.2.4 *Control variables*

In our analyses, we controlled for several tenant and SP characteristics. We controlled for firm age and firm size as both variables may affect the extent to which tenants benefit from their SP residence (van der Borgh et al., 2012; Vásquez-Urriago et al., 2016a). Age is calculated as the number of years since the firm was legally established and is derived from the Orbis Europe database. Size was measured by asking the respondents the number of employees of the firm on the SP (in full-time equivalents, at the end of 2017). Further, we controlled for sector by introducing four dummy variables: information technology and communication ("Sector IT"), life sciences and chemical ("Sector Life sciences"), new materials and engineering ("Sector Material"), and other sectors ("Sector other") (Díez-Vial & Montoro-Sánchez, 2016). The sector 'life sciences and chemical' served as reference category. Since our study is conducted in three countries, and as geographical characteristics may affect organizational sponsorship (Amezcua et al., 2013), we incorporated dummy variables to account for geographic differences: "Belgium" and "Spain", with "Denmark" as reference category.

As it is acknowledged that SPs are heterogeneous (Link & Link, 2003; Siegel et al., 2003a) and that SP characteristics also affect the benefits tenants may gain from SP residence (Albahari et al., 2018), we also included SP age and SP size as control variables in this study. These variables are derived from face-to-face interviews we had with the SP managers and complemented with information we found on the SP websites. SP age is measured as the number of years since the legal establishment of the park, with a mean of 25.99 years (s.d. 15.21). We finally incorporated SP size as a control variable, measured by the total number of SP tenants. The average SP size for the tenants in our sample is 88.97 (s.d. 39.19).

4 Results

Structural equation modeling (SEM) is used to test our conceptual model, thereby following the well-established two-stage procedure (Anderson & Gerbing, 1988). First, we estimate the measurement model to test whether our constructs show sufficient reliability and validity. Second, the structural model allows for testing the hypothesized relationships. We employed the 'Lavaan', 'Mediation', and 'ggplot2' package in R (respectively, Rosseel, 2012; Tingley et al., 2014; Wickham, 2011).

4.1 Measurement model

Table 3 summarizes the CFA for our measurement model. Particularly, we present the items for each construct, the standardized factor loadings, the average variance extracted, and the composite reliability. All factor loadings are statistically significant (Anderson & Gerbing, 1988) and range between 0.50 and 0.88, which indicates good convergent validity as the general recommended minimum is 0.5 (Hair et al., 2014) and even 0.4 in the social sciences (Yli-Renko et al., 2001).[5] We also calculated the average variance extracted (AVE) for each latent construct. Typically, an AVE of 0.5 and higher suggests adequate convergence (Hair et al., 2014). While one construct (i.e., 'Use intensity of buffering services') is slightly below this threshold, the others meet this criterion. Finally, the composite reliabilities (CR) of all four constructs are well above the recommended minimum of 0.6 (Bagozzi & Yi, 1988). We can conclude that our measurement model performs well in terms of convergent validity.

Further, we tested the discriminant validity of our measurement model. Particularly, we compared

[5] Note that items 'FE4' and 'FE5' have been removed from the analysis as they did not meet these criteria.

Table 3 Results of confirmatory factor analysis

Latent variable	Items	Standardized factor loadings*	Average variance extracted (AVE)	Composite reliability
TMT functional experience	FE1	0.87	0.50	0.79
	FE2	0.68		
	FE3	0.58		
	FE6	0.64		
Use intensity of buffering services	US_prop	0.57	0.45	0.76
	US_bus	0.71		
	US_innov	0.62		
	US_netw	0.76		
Boundary-spanning capacity	BS1	0.71	0.60	0.86
	BS2	0.80		
	BS3	0.82		
	BS4	0.76		

*$p < 0.001$ for all standardized factor loadings

the AVE for any two constructs with the square of the correlation estimate between these two constructs (Fornell & Larcker, 1981). We examined all constructs and found no problems in terms of discriminant validity. Additionally, we conducted the chi-square test for discriminant validity (Anderson & Gerbing, 1988; Hair et al., 2014). Particularly, we tested whether the fit of our four-factor model is significantly different from any possible three-factor model, two-factor model, or the one-factor model. The results show that the chi-square in our four-factor model is significantly lower than in the other models. We conclude that our constructs are unique and distinct from each other.

As stated by Hair et al. (2014), many indices can be used to examine the fit of a measurement model. We report the comparative fit index (CFI), the Tucker–Lewis index (TLI), the root mean square error of approximation (RMSEA), the standardized root mean residual (SRMR), and the chi-square value with the degrees of freedom, as each of these indicators provides information about specific aspects of model fit (Kollmann & Stöckmann, 2014). Values of 0.90 or higher are considered to indicate adequate fit for CFI and TLI (Hair et al., 2014; Hu & Bentler, 1998). Values up to 0.08 are acceptable for RMSEA (Bagozzi & Yi, 1988) and values up to 0.10 for SRMR (Kollmann & Stöckmann, 2014; Weston & Gore, 2006). Our measurement model fits the data well (CFI = 0.984, TLI = 0.980, RMSEA = 0.032, SRMR = 0.048, chi-square: 61.099 with $p = 0.16$ and df = 51).

As the latent constructs in our model are collected in the same survey, we verified whether common method variance could have biased our findings. First, we conducted the Harman's single-factor test (Podsakoff & Organ, 1986), conducting a PCA on all items to test for common method variance. If one single factor emerges or one general factor accounts for most of the covariance in the variables, common method variance poses a problem (Podsakoff et al., 2003). The results indicate that four factors emerge and none of these factors account for the majority of the variance. Second, we also tested for common method variance by conducting a CFA. If common method variance posed a problem, a model in which all variables are allowed to load onto a single factor would fit the data as well as a more complex model (e.g., Hoobler et al., 2009). This model however demonstrates very poor fit (Chi-square: 462.93 with $p < 0.001$ and df = 54, CFI = 0.47, TLI = 0.36, RMSEA = 0.20, SRMR = 0.18). Third, we followed the instructions by Williams et al., (2010) and used confirmatory factor analysis with a marker variable. A marker variable needs to be uncorrelated to the substantial variables (TMT functional experience, use intensity of buffering services, and boundary-spanning capacity). We chose TMT trust as a marker variable (measured as a 5-item scale in De Jong et al. (2010)). TMT trust is unrelated to the substantial variables as the correlations indicate (−0.053 for TMT functional experience, −0.067 for use intensity of buffering services, 0.109 for boundary-spanning capacity; all p

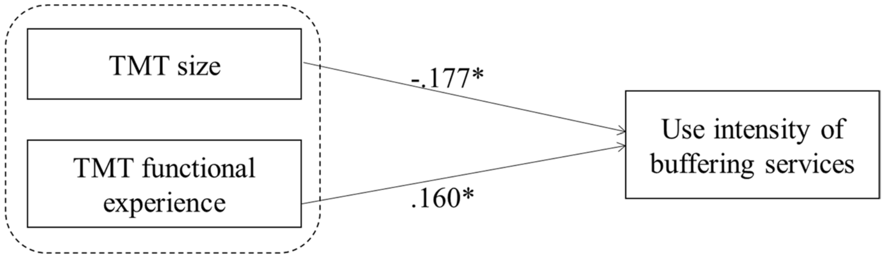

Fig. 2 Results of the structural equation model (model 1). Significance level: *$p<0.05$, **$p<.01$, ***$p<0.001$; $n=201$. Note: All control variables are included yet not displayed

values > 0.05). The method proposed by Williams et al., (2010) provides a technique to quantify the amount of method variance associated with the measurement of latent variables. Specifically, it relies on the decomposition of a reliability measure into substantive and method variance components in order to assess the impact of method variance on latent variable measurement. We decomposed the reliability measure for our three substantive variables, and found that method components only accounted for a small percentage of the reliability values for these substantive variables (11.22% for TMT functional experience, 3.10% for use intensity of buffering service, and 3.34% for boundary-spanning capacity). Furthermore, chi-square model comparison tests determined that the effects of the marker variable did not significantly bias factor correlation estimates.[6] Taken together, these tests indicate that common method variance is unlikely to affect our results.

4.2 Structural model

In the next step of our analyses, we test our hypothesized relationships between the variables. While Fig. 2 presents the standardized parameter estimates and related significance of our structural model without the interaction effects (model 1), Fig. 3 shows the estimates of the model with interaction effects (model 2). The findings are summarized in Table 4.

Hypothesis 1a suggests a negative relationship between TMT size and the use intensity of buffering services. The results of model 1 confirm this negative relationship ($\beta=-0.177$, $p<0.05$). Hypothesis 1a is supported. Similarly, hypothesis 1b posits a negative relationship between the functional experience of a tenant's TMT and the use intensity of buffering services. However, as we find TMT functional experience to be positively related to the use of SP services ($\beta=0.160$, $p<0.05$), we reject hypothesis 1b.

Subsequently, hypotheses 2a and 2b argue that the relationships between TMT size and TMT functional experience on the one hand, and the use intensity of buffering services on the other hand, are negatively moderated by tenant boundary-spanning capacity. As model 2 shows, this moderating effect is statistically significant for the relationship between TMT size and the use intensity of buffering services ($\beta=-0.162$,

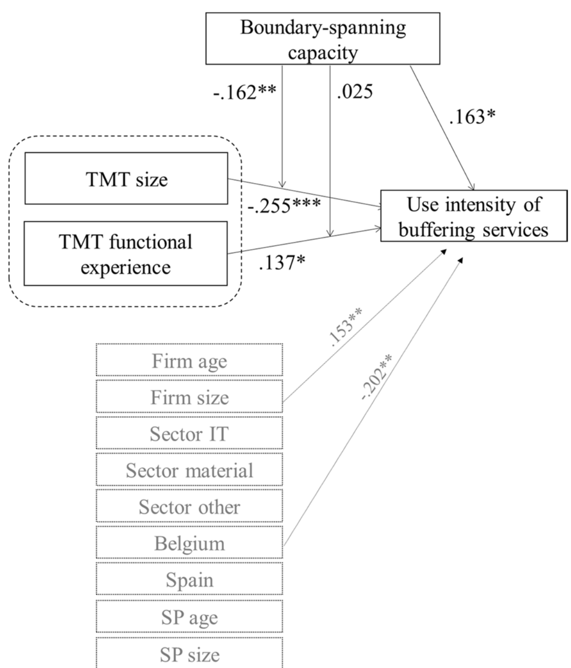

Fig. 3 Results of the structural equation model (model 2). Significance level: *$p<0.05$, **$p<0.01$, ***$p<0.001$; $n=201$

[6] The difference between the baseline model and the method-C model is not statistically significant ($p=.95$). Neither is the difference between the method-C and method-U model ($p=1$), nor the difference between the method-U and method-R model ($p=0.68$).

Table 4 Results of the structural equation model

Structural path	Standardized coefficient (β)	s.e	*z* value	*p* value	Conclusion
Hypotheses					
Model 1 (only mediation)					
TMT size → use intensity of buffering services	−0.177	0.043	−2.524	0.012	H1a supported
TMT functional experience → use intensity of buffering services	0.160	0.069	2.341	0.019	H1b rejected
Model 2 (with interaction)					
TMT size * boundary-spanning capacity → use intensity of buffering services	−0.162	0.018	-2.953	0.003	H2a supported
TMT functional experience * boundary-spanning cap. → use intensity of buffering services	0.025	0.061	0.316	0.752	H2b not supported

Maximum likelihood estimation with robust (Huber–White) standard errors, $n = 201$

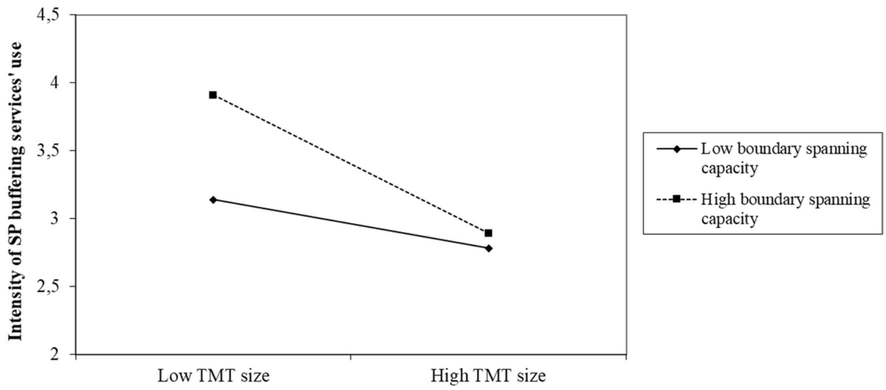

Fig. 4 Moderation effect of boundary-spanning capacity on the relationship between TMT size and the use intensity of buffering services

$p < 0.01$). Hypothesis 2a is therefore supported while hypothesis 2b is not. In Fig. 4, we plot the interaction effect of boundary-spanning capacity on the relationship between TMT size and the use intensity of buffering services (hypothesis 2a). Figure 4 indicates that tenants with a small TMT (mean–s.d.) will use more services, especially if they have high levels of boundary-spanning capacity (mean + s.d.). This effect is less pronounced for tenants with larger TMTs (mean + s.d.).

We further investigate this interaction effect by plotting the marginal effects of TMT size on the use intensity of buffering services at increasing levels of tenant boundary-spanning capacity. Particularly, Fig. 5 depicts the two-way interaction between TMT size and boundary-spanning capacity predicting the use intensity of buffering services. We plotted the simple slopes for the relation between TMT size and the use intensity of buffering services on the *y*-axis, with higher values representing a stronger and more positive relation between TMT size and use intensity. Boundary-spanning capacity is plotted on the *x*-axis. Following recent examples, we standardized our moderator variable to make this plot (Barstead et al., 2018). The solid black line represents the model-estimated simple slope at increasing levels of boundary-spanning capacity. The shaded region marks the upper and lower boundaries of the 95% confidence interval for the model-estimated simple slopes, and the horizontal dotted line indicates a simple slope of zero, providing a visual indicator for determining whether an estimated slope at a given value of boundary-spanning capacity is significantly different from zero. The figure indicates that boundary-spanning capacity attenuates the association between TMT size and the use intensity of buffering services.

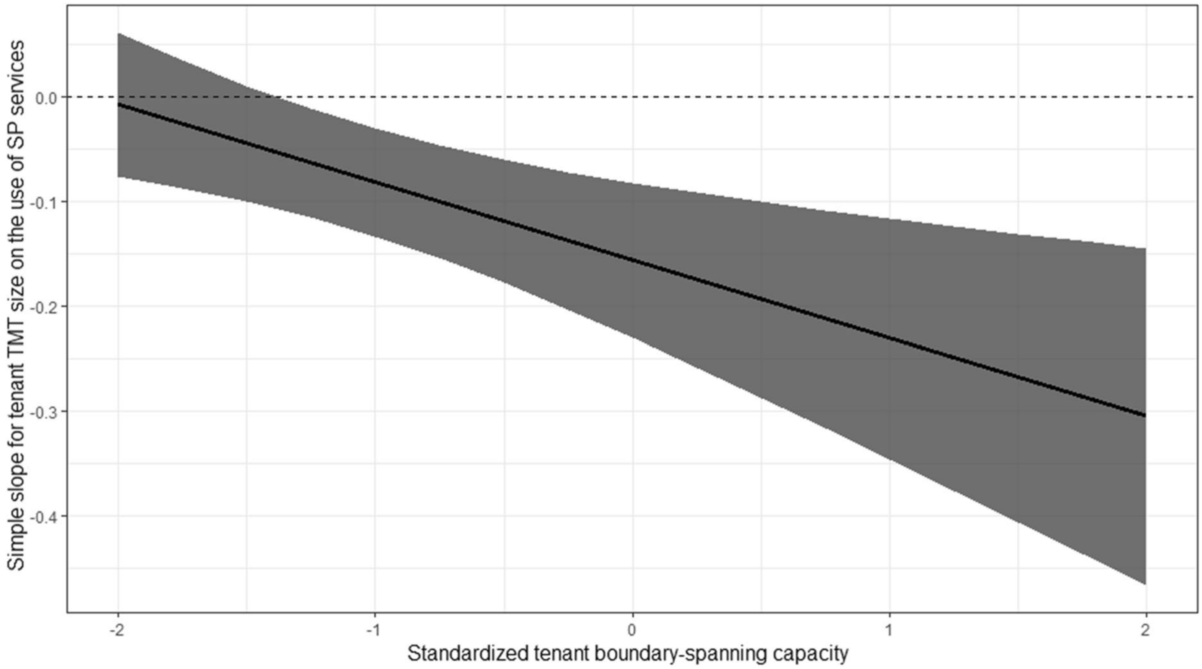

Fig. 5 Marginal effect of TMT size on the use intensity of buffering services at increasing levels of boundary-spanning capacity

As boundary-spanning capacity increases, the association between TMT size and the use intensity of buffering services is increasingly negative. Particularly, TMT size significantly predicts the use intensity of services at values of boundary-spanning capacity greater than about -1.4 (standardized value).

As to what the control variables are concerned, we find a significantly positive relationship between firm size and the use intensity of buffering services ($\beta = 0.153$, $p < 0.01$). The country dummy 'Belgium' is negatively related to the use intensity of SP buffering services compared to its reference category 'Denmark' ($\beta = -0.202$, $p < 0.01$).

4.3 Post hoc analyses

We carried out several post hoc analyses to provide more fine-grained insights into our findings. First, we investigated whether the two independent TMT variables interact with each other in explaining the use intensity of buffering services. The analyses indicated that the interaction effect between TMT size and functional experience is statistically insignificant. Second, in order to provide deeper insights into the use intensity of specific buffering services, we reran model 1 by using the different items of the use intensity of buffering services. The results indicate that TMT size is significantly negatively related to the use intensity of business support and innovation services, whereas TMT functional experience is positively related to the use of property-based and networking services, which provides some insights into the specific services that are used to a greater or lesser extent depending on the firm's managerial resources. Third, as it is reasonable that the extent to which tenants use SP buffering services also depends on the amount of services SPs offer, we reran our models including an additional control variable which represents the number of services SPs offered on the SP. This variable was derived from the face-to-face interviews we had with the SP managers during which we presented them with a list of services and facilities based upon extant literature (e.g., Chan & Lau, 2005; Westhead & Batstone, 1998) and asked them to indicate which were provided on park. We reran model 1, hereby incorporating the number of SP services offered as a control variable, and while we found that there is a significantly positive relationship between the number of services offered on SP and the use intensity of buffering services

($\beta = 0.618$, $p < 0.001$), our results and conclusions regarding our hypotheses remained robust.

5 Discussion

Our study started from the premise that it is not because organizational sponsorship initiatives are offered that such initiatives are utilized to the same extent by all firms who could benefit from them. Although some scholars have recently hinted that firms may differ with regard to which service offerings they utilize (Breivik-Meyer et al., 2020; Van Weele et al., 2017), the explanations offered in extant literature are mostly exploratory and center on the organizational sponsorship initiative, pointing at either low quality of the offerings (e.g., Rice, 2002) or a mismatch between the offerings and the needs of the tenant base (e.g., Bruneel et al., 2012). Our findings confirm the importance of studying use intensity in the context of organizational sponsorship. Our resource dependence lens and focus on firm-level characteristics, particularly managerial resources, proved to be pertinent as we found that the use intensity of buffering services is largely contingent on such characteristics. The effect of TMT resources on the use intensity of buffering services was further moderated by TMT boundary-spanning capacity. Contradictory to our expectations, we detected a positive relationship between TMT functional experience and the use intensity of SP buffering services. We believe this unexpected finding can be explained from an absorptive capacity or human capital perspective: tenants with more experienced TMTs may have learned about the importance of buffering their resource dependence by using services at hand, and may therefore use SP services to a larger extent. Alternatively, highly diverse teams may struggle to coordinate their activities (Sivasubramaniam et al., 2012), and may therefore more frequently call upon the SP for support. In what follows, we discuss the contributions to the existing literature, the limitations, and the practical implications of this research.

5.1 Implications

First, our study contributes to an ongoing debate on the boundary conditions of organizational sponsorship (Amezcua et al., 2020). While extant studies have focused on understanding whether or not organizational sponsorship initiatives enhance firm performance and survival (Amezcua et al., 2013), most studies have treated reliance on such initiatives as a dummy variable, typically indicating whether or not a firm is located in an incubator or on a science park. As hinted at by scholars recently (Breivik-Meyer et al., 2020; Van Weele et al., 2017), our study confirms that firms may differ in regard to which service offerings they may utilize. As such, while prior research has documented that the effectiveness of organizational sponsorship is likely affected by environmental characteristics, such as geographic-based founding density (Amezcua et al., 2013), local economic conditions (Amezcua et al., 2020), external technical, economic and cultural factors (Marquis & Huang, 2009), or the nature of the organizational sponsorship initiative (Schwartz & Hornych, 2010), our study complements these insights by pointing to the importance of considering the extent to which services offered by organizational sponsorship initiatives are actually used (and not only offered). As such, it indicates that future studies into organizational sponsorship may benefit from moving beyond the mere assessment of whether a sponsorship initiative is in place, or a service being offered, to the integration of actual sponsorship (services') use. Additionally, our study points to an important firm-level contingency factor, namely managerial resources, to be considered as explanatory or control variable in future studies into the impact of organizational sponsorship in general, and the buffering mechanism specifically.

Second, this study contributes to resource dependence theory by pointing to the reliance on organizational sponsorship initiatives as a means for firms to overcome their dependence on the environment. It further contributions by indicating under which firm-level, and particularly TMT-specific characteristics, this is more or less the case. As such, while organizational sponsorship initiatives may be a means of increasing resource munificence and minimizing environmental resource dependence for firms (Amezcua et al., 2013; Astley and Van de Ven, 1983; Pfeffer & Salancik, 1978), our study indicates that the extent to which this occurs is likely contingent on firm-specific factors. As such, future studies into firm resource dependence may purposefully consider such factors as explanatory or control variables.

Finally, our study contributes to the science park literature. Specifically, our study contributes by pointing to a potentially important building block for understanding SP effectiveness, namely the extent to which SP buffering services are not only offered by SPs, but also actively used by potential beneficiary firms. As such, we have challenged the taken-for-granted assumption that SPs invariably contribute to all tenants as they offer services. Specifically, our study complements work in the field which has examined boundary conditions at the level of the science park, such as SP management (e.g., Ng et al., 2021), SP size (e.g., Gwebu et al., 2019) and SP design (e.g., Ng et al., 2019), by pointing to boundary conditions at the level of the firm. As such, our study calls for future research into the role of SPs to incorporate the use of SP services as an explanatory, mediator, or dependent variable.

Our study has implications for (potential) tenants, SP managers, and policymakers. Policymakers have recently shown much interest in gaining a better understanding of the conditions under which organizational sponsorship initiatives may help the firms for which they were designed (Rowe, 2014). Our study thus contributes by shedding light on which firms are more or less likely to rely on the (buffering) services offered by organizational sponsorship initiatives. By consequence, also SP managers could a priori evaluate which type of tenants will be able to gain most from the buffering services they offer. At the same time, our insights inform aspiring SP tenants who may be concerned with understanding how organizational sponsorship (and specifically: SP) initiatives may help through services, by pointing to firm-level characteristics which may have them call upon buffering services to a greater or lesser extent in order to overcome resource dependence. Our findings also suggest that it is essential for SP managers to communicate the availability of SP services to the tenants and promote the use of such services. While the former recommendation may seem evident, throughout our interviews, it became apparent that there were many discrepancies between what the SP offered and what the tenants thought the SP offered.

5.2 Limitations and future research directions

Despite these contributions and implications, this research is subject to some limitations that represent opportunities for future research. First, we pointed out that important conditions exist that affect the intensity with which services offered by organizational sponsorship initiatives are utilized. By building on resource dependence theory, we have specifically focused on the influence of human resource-related tenant conditions, i.e., managerial resources and boundary-spanning capacity. However, we encourage future research to focus on other firm-level boundary conditions. Future research could investigate other internal resources, such as financial resources (Wiklund et al., 2010), and the extent to which they affect firms' reliance on organizational sponsorship services. Alternatively, future research could study the interplay of boundary conditions at different levels of analysis, combining contingencies at the level of the firm, the organizational sponsorship initiative, and the environment. Second, we conducted our study in a particular region of the world, namely in diverse countries in Europe. Future research could purposefully assess whether our findings are also applicable in other regions of the world, or whether differences exist across institutional contexts. Third, in this paper, we deliberately focused on boundary conditions for the use intensity of services offered by science parks. We consider this an important step forward as prior research has mainly considered the reliance on such initiatives or their services as dummy variables—indicating whether or not organizational sponsorship is offered. Our study thus indicates that it is not because an initiative exists that it is by definition used (to the same extent) by potential beneficiaries. Future research could build upon this important insight, and use a longitudinal research design to investigate the relationship between the use intensity of (buffering) services and its impact on firm performance and survival. Alternatively, it could study other organizational sponsorship initiatives, which may offer access to alternative resources than science parks, and investigate whether our results also hold for such alternative resource offerings.

Appendix 1

Table 5 Latent variables, concrete items, and mean per item

TMT functional experience		Mean (s.d.)
Please rate the degree to which the members of the firm's TMT have experience in the following domains		
FE1	Management	5.58 (1.64)
FE2	Marketing, sales and promotion	4.70 (1.90)
FE3	Accounting, controlling and financing	4.68 (1.81)
FE4	Engineering and R&D	5.83 (1.63)
FE5	Production	4.55 (1.80)
FE6	Personnel	5.06 (1.50)
1 = to a low degree, 7 = to a high degree		
Use intensity of buffering services		
Since your firm established on the science park, to which extent have you used …		
US_prop	… Property-related services on park? (e.g., conference and meeting rooms, secretarial services, cafeteria and restaurant, maintenance, recreational facilities)	5.00 (1.97)
US_bus	… Business support services offered/initiated by the science park management and staff? (e.g., business advice, financial/marketing/legal advice, bookkeeping services, mentoring, business-related training sessions)	2.45 (1.78)
US_innov	… Innovation services offered/initiated by the science park management and staff? (e.g., R&D services, specialized laboratories, scientific equipment, advice on R&D and technology, technology transfer services)	2.15 (1.58)
US_netw	… Networking services offered/initiated by the science park management and staff? (e.g., networking and social events that give access to potential customers, suppliers, funding sources, partners within or beyond the science park)	3.42 (1.75)
1 = to a very little extent, 7 = to a very large extent		
Boundary-spanning capacity		
To which extent does …		
BS1	… The firm encourages its members to solicit information and resources from elsewhere?	5.17 (1.53)
BS2	… The firm encourages its members to try to influence important actors elsewhere on behalf of the firm and its work?	4.63 (1.64)
BS3	… The firm values its members for making use of their relationships with others on behalf of the firm?	5.06 (1.67)
BS4	… The work of the firm depends upon information and resources actively solicited by firm members, that is, information and resources beyond what comes through official channels?	4.44 (1.75)
1 = to a very little extent, 7 = to a very large extent		

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References

Albahari, A., Barge-Gil, A., Pérez-Canto, S., & Modrego, A. (2018). The influence of science and technology park characteristics on firms' innovation results. *Papers in Regional Science, 97*(2), 253–279. https://doi.org/10.1111/pirs.12253

Aldrich, H. E., & Fiol, C. M. (1994). Fools rush in? The institutional context of industry creation. *Academy of Management Review, 19*(4), 645–670. https://doi.org/10.5465/amr.1994.9412190214

Amason, A. C. (1996). Distinguishing the effects of functional and dysfunctional conflict on strategic decision making: Resolving a paradox for top management teams. *Academy of Management Journal, 39*(1), 123–148. https://doi.org/10.5465/256633

Amason, A. C., & Sapienza, H. J. (1997). The effects of top management team size and interaction norms on cognitive and affective conflict. *Journal of Management, 23*(4), 495–516. https://doi.org/10.1177/014920639702300401

Amezcua, A. S., Grimes, M. G., Bradley, S. W., & Wiklund, J. (2013). Organizational sponsorship and founding environments: A contingency view on the survival of business-incubated firms, 1994–2007. *Academy of Management Journal, 56*(6), 1628–1654. https://doi.org/10.5465/amj.2011.0652

Amezcua, A., Ratinho, T., Plummer, L., & Jayamohan, P. (2020). Organizational sponsorship and the economies of place: How regional urbanization and localization shape incubator outcomes. *Journal of Business Venturing, 35*(4). https://doi.org/10.1016/j.jbusvent.2019.105967

Anderson, J., & Gerbing, D. (1988). Structural equation modeling in practice: A review and recommended two-step approach. *Psychological Bulletin, 103*(3), 411–423. https://doi.org/10.1037/0033-2909.103.3.411

Armstrong, J. S., & Overton, T. S. (1977). Estimating non-response bias in mail surveys. *Journal of Marketing Research,* 396-402.https://doi.org/10.1177/002224377701400320

Astley, W. G., & Van de Ven, A. H. (1983). Central perspectives and debates in organization theory. *Administrative Science Quarterly,* 245-273. https://doi.org/10.2307/2392620

Audretsch, D. B. (1991). New-firm survival and the technological regime. *The Review of Economics and Statistics, 73*(3), 441–450. https://doi.org/10.2307/2109568

Autio, E., & Rannikko, H. (2016). Retaining winners: Can policy boost high-growth entrepreneurship? *Research Policy, 45*(1), 42–55. https://doi.org/10.1016/j.respol.2015.06.002

Bagozzi, R., & Yi, Y. (1988). On the evaluation of structural equation models. *Journal of the Academy of Marketing Science, 16*(1), 74–94. https://doi.org/10.1177/009207038801600107

Bantel, K. A., & Jackson, S. E. (1989). Top management and innovations in banking: Does the composition of the top team make a difference? *Strategic Management Journal, 10*(S1), 107–124. https://doi.org/10.1002/smj.4250100709

Barstead, M. G., Smith, K. A., Laursen, B., Booth-LaForce, C., King, S., & Rubin, K. H. (2018). Shyness, preference for solitude, and adolescent internalizing: The roles of maternal, paternal, and best-friend support. *Journal of Research on Adolescence, 28*(2), 488–504. https://doi.org/10.1111/jora.12350

Benneworth, P., & Ratinho, T. (2014). Reframing the role of knowledge parks and science cities in knowledge-based urban development. *Environment and Planning c: Government and Policy, 32*(5), 784–808. https://doi.org/10.1068/c1266r

Bøllingtoft, A. (2012). The bottom-up business incubator: Leverage to networking and cooperation practices in a self-generated, entrepreneurial-enabled environment. *Technovation, 32*(5), 304–315. https://doi.org/10.1016/j.technovation.2011.11.005

Breivik-Meyer, M., Arntzen-Nordqvist, M., & Alsos, G. (2020). The role of incubator support in new firms accumulation of resources and capabilities. *Innovation Organization and Management, 3*, 228–249. https://doi.org/10.1080/14479338.2019.1684204

Bruneel, J., Rathinho, T., Clarysse, B., & Groen, A. (2012). The evolution of business incubators: Comparing demand and supply of business incubation services across different incubator generations. *Technovation, 32*(2), 110–121. https://doi.org/10.1016/j.technovation.2011.11.003

Bunderson, J. S., & Sutcliffe, K. M. (2002). Comparing alternative conceptualizations of functional diversity in management teams: Process and performance effects. *Academy of Management Journal, 45*(5), 875–893.

Cantner, U., Goethner, M., & Stuetzer, M. (2010). Disentangling the effects of new venture team functional heterogeneity on new venture performance. *Jena Economic Research Papers no.* 2010–2029. Jena, Germany: Max Planck Institute of Economics.

Chan, K. F., & Lau, T. (2005). Assessing technology incubator programs in the science park: The good, the bad and the ugly. *Technovation, 25*(10), 1215–1228. https://doi.org/10.1016/j.technovation.2004.03.010

Colombo, M. G., & Delmastro, M. (2002). How effective are technology incubators?: Evidence from Italy. *Research Policy, 31*(7), 1103–1122. https://doi.org/10.1016/S0048-7333(01)00178-0

Daellenbach, U. S., McCarthy, A. M., & Schoenecker, T. S. (1999). Commitment to innovation: The impact of top management team characteristics. *R&D Management, 29*(3), 199–208. https://doi.org/10.1111/1467-9310.00130

Day, D. V., & Lord, R. G. (1992). Expertise and problem categorization: The role of expert processing in organizational sense-making. *Journal of Management Studies, 29*(1), 35–47.

De Massis, A., Audretsch, D., Uhlaner, L., & Kammerlander, N. (2018). Innovation with limited resources:

Management lessons from the German Mittelstand. *Journal of Product Innovation Management, 35*(1), 125–146. https://doi.org/10.1111/jpim.12373

De Jong, B. A., & Elfring, T. (2010). How does trust affect the performance of ongoing teams? The mediating role of reflexivity, monitoring, and effort. *Academy of Management Journal, 53*(3), 535–549. https://doi.org/10.5465/AMJ.2010.51468649

Dettwiler, P., Lindelöf, P., & Löfsten, H. (2006). Utility of location: A comparative survey between small new technology-based firms located on and off Science Parks—Implications for facilities management. *Technovation, 26*(4), 506–517.

Díez-Vial, I., & Fernández-Olmos, M. (2017). The effect of science and technology parks on firms' performance: How can firms benefit most under economic downturns? *Technology Analysis & Strategic Management, 29*(10), 1153–1166. https://doi.org/10.1080/09537325.2016.1274390

Díez-Vial, I., & Montoro-Sánchez, Á. (2016). How knowledge links with universities may foster innovation: The case of a science park. *Technovation, 50–51*, 41–52. https://doi.org/10.1016/j.technovation.2015.09.001

Díez-Vial, I., & Montoro-Sánchez, Á. N. (2017). From incubation to maturity inside parks: The evolution of local knowledge networks. *International Journal of Technology Management, 73*(1–3), 132–150. https://doi.org/10.1504/ijtm.2017.10003244

Faraj, S., & Yan, A. (2009). Boundary work in knowledge teams. *Journal of Applied Psychology, 94*(3), 604–617. https://doi.org/10.1037/a0014367

Felsenstein, D. (1994). University-related science parks — 'seedbeds' or 'enclaves' of innovation? *Technovation, 14*(2), 93–110. https://doi.org/10.1016/0166-4972(94)90099-X

Finkelstein, S., & Hambrick, D. C. (1996). *Strategic leadership: Top executives and their effects on organizations*. West.

Flynn, D. (1993). A critical exploration of sponsorship, infrastructure, and new organizations. *Small Business Economics, 5*(2), 129–156. https://doi.org/10.1007/BF01531911

Fornell, C., & Larcker, D. F. (1981). Structural equation models with unobservable variables and measurement error: Algebra and statistics. *Journal of Marketing Research, 18*(3), 382–388. https://doi.org/10.1177/002224378101800313

Guy, I. (1996). A look at Aston Science Park. *Technovation, 16*(5), 217–218.

Gwebu, K. L., Sohl, J., & Wang, J. (2019). Differential performance of science park firms: An integrative model. *Small Business Economics, 52*(1), 193–211. https://doi.org/10.1007/s11187-018-0025-5

Hair, J., Black, W., Babin, B., & Anderson, R. (2014). *Multivariate Data Analysis*. Pearson Education Limited, Essex.

Hall, R. (1982). *Organizations: Structure and process* (3rd ed.). Prentice Hall.

Hambrick, D. C., & D'Aveni, R. A. (1992). Top team deterioration as part of the downward spiral of large corporate bankruptcies. *Management Science, 38*(10), 1445–1466. https://doi.org/10.1287/mnsc.38.10.1445

Helfat, C. E., & Peteraf, M. A. (2015). Managerial cognitive capabilities and the microfoundations of dynamic capabilities. *Strategic Management Journal, 36*(6), 831–850. https://doi.org/10.1002/smj.2247

Hoobler, J. M., Wayne, S. J., & Lemmon, G. (2009). Bosses' perceptions of family-work conflict and women's promotability: Glass ceiling effects. *Academy of Management Journal, 52*(5), 939–957. https://doi.org/10.5465/amj.2009.44633700

Hu, L., & Bentler, P. (1998). Fit indices in covariance structure modeling: Sensitivity to underparameterized model misspecification. *Psychological Methods, 3*(4), 424–453. https://doi.org/10.1037/1082-989X.3.4.424

Huse, M. (2005). Accountability and creating accountability: A framework for exploring behavioural perspectives of corporate governance. *British Journal of Management, 16*, S65–S79.

IASP. (2020). About IASP. Retrieved October 10, 2020 from https://www.iasp.ws/about-us/About-IASP.

Jourdan, J., & Kivleniece, I. (2017). Too much of a good thing? The dual effect of public sponsorship on organizational performance. *Academy of Management Journal, 60*(1), 55–77. https://doi.org/10.5465/amj.2014.1007

Kihlgren, A. (2003). Promotion of innovation activity in Russia through the creation of science parks: The case of St Petersburg (1992–1998). *Technovation, 23*(1), 65–76. https://doi.org/10.1016/S0166-4972(01)00077-3

Klofsten, M., & Jones-Evans, D. (1996). Stimulation of technology-based small firms—A case study of university-industry cooperation. *Technovation, 16*(4), 187–213. https://doi.org/10.1016/0166-4972(95)00052-6

Knockaert, M., Bjornali, E. S., & Erikson, T. (2015). Joining forces: Top management team and board chair characteristics as antecedents of board service involvement. *Journal of Business Venturing, 30*(3), 420–435. https://doi.org/10.1016/j.jbusvent.2014.05.001

Koçak, Ö., & Can, Ö. (2014). Determinants of inter-firm networks among tenants of science technology parks. *Industrial and Corporate Change, 23*(2), 467–492. https://doi.org/10.1093/icc/dtt015

Kollmann, T., & Stöckmann, C. (2014). Filling the entrepreneurial orientation–performance gap: The mediating effects of exploratory and exploitative innovations. *Entrepreneurship Theory and Practice, 38*(5), 1001–1026. https://doi.org/10.1111/j.1540-6520.2012.00530.x

Kor, Y. Y. (2003). Experience-based top management team competence and sustained growth. *Organization Science, 14*(6), 707–719. https://doi.org/10.1287/orsc.14.6.707.24867

Laursen, K., & Salter, A. (2006). Open for innovation: The role of openness in explaining innovation performance among UK manufacturing firms. *Strategic Management Journal, 27*(2), 131–150. https://doi.org/10.1002/smj.507

Lazzarini, S. G. (2015). Strategizing by the government: Can industrial policy create firm-level competitive advantage? *Strategic Management Journal, 36*(1), 97–112. https://doi.org/10.1002/smj.2204

Levina, N., & Vaast, E. (2005). The emergence of boundary spanning competence in practice: Implications for implementation and use of information systems. *MIS Quarterly,* 335-363https://doi.org/10.2307/25148682

Lindelöf, P., & Löfsten, H. (2003). Science park location and new technology-based firms in Sweden–implications for strategy and performance. *Small Business Economics, 20*(3), 245–258. https://doi.org/10.1023/A:1022861823493

Link, A. N., & Link, K. R. (2003). On the growth of US Science Parks. *The Journal of Technology Transfer, 28*(1), 81–85. https://doi.org/10.1023/A:1021634904546

Link, A. N., & Scott, J. T. (2003). The growth of Research Triangle Park. *Small Business Economics, 20*(2), 167–175.

Löfsten, H., & Lindelöf, P. (2002). Science parks and the growth of new technology-based firms—Academic-industry links, innovation and markets. *Research Policy, 31*(6), 859–876. https://doi.org/10.1016/S0048-7333(01)00153-6

Löfsten, H., & Lindelöf, P. (2003). Determinants for an entrepreneurial milieu: Science parks and business policy in growing firms. *Technovation, 23*(1), 51–64. https://doi.org/10.1016/S0166-4972(01)00086-4

Lynn, M. L. (2005). Organizational buffering: Managing boundaries and cores. *Organization Studies, 26*(1), 37–61.

Marquis, C., & Huang, Z. (2009). The contingent nature of public policy and the growth of U.S. commercial banking. *Academy of Management Journal, 52*, 1222–1246. https://doi.org/10.5465/amj.2009.47084828

McAdam, M., & McAdam, R. (2008). High tech start-ups in University Science Park incubators: The relationship between the start-up's lifecycle progression and use of the incubator's resources. *Technovation, 28*(5), 277–290. https://doi.org/10.1016/j.technovation.2007.07.012

Menz, M. (2012). Functional top management team members: A review, synthesis, and research agenda. *Journal of Management, 38*(1), 45–80.

Ng, W. K. B., Appel-Meulenbroek, R., Cloodt, M., & Arentze, T. (2019). Towards a segmentation of science parks: A typology study on science parks in Europe. *Research Policy, 48*(3), 719–732. https://doi.org/10.1016/j.respol.2018.11.004

Ng, W. K. B., Appel-Meulenbroek, R., Cloodt, M., & Arentze, T. (2021). Perceptual measures of science parks: Tenant firms' associations between science park attributes and benefits. *Technological Forecasting and Social Change, 163*, 120408. https://doi.org/10.1016/j.techfore.2020.120408

Pfeffer, J., & Salancik, G. R. (1978). *The external control of organizations: A resource dependence perspective*. Harper & Row. https://doi.org/10.2307/2231527

Phan, P. H., Siegel, D. S., & Wright, M. (2005). Science parks and incubators: Observations, synthesis and future research. *Journal of Business Venturing, 20*(2), 165–182. https://doi.org/10.1016/j.jbusvent.2003.12.001

Podsakoff, P., & Organ, D. (1986). Self-reports in organizational research: Problems and prospects. *Journal of Management, 12*(4), 531–544. https://doi.org/10.1177/014920638601200408

Podsakoff, P., MacKenzie, S., Lee, J.-Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology, 88*(5), 879–903. https://doi.org/10.1037/0021-9010.88.5.879

Radosevic, S., & Myrzakhmet, M. (2009). Between vision and reality: Promoting innovation through technoparks in an emerging economy. *Technovation, 29*(10), 645–656. https://doi.org/10.1016/j.technovation.2009.04.001

Rice, M. (2002). Co-production of business assistance in business incubators: An exploratory study. *Journal of Business Venturing, 17*(2), 163–187. https://doi.org/10.1016/S0883-9026(00)00055-0

Rosenkopf, L., & Nerkar, A. (2001). Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal, 22*(4), 287–306. https://doi.org/10.1002/smj.160

Rosseel, Y. (2012). Lavaan: An R package for structural equation modeling. *Journal of Statistical Software, 48*(2), 1–36. http://hdl.handle.net/1854/LU-3099674.

Rowe, D. (2014). *Setting up, Managing and Evaluating EU Science and Technology parks - An advice and guidance report on good practice*. Brussels, Belgium: European Commission, Directorate-General for Regional and Urban policy.

Salvador, E. (2011). Are science parks and incubators good "brand names" for spin-offs? The case study of Turin. *The Journal of Technology Transfer, 36*(2), 203–232. https://doi.org/10.1007/s10961-010-9152-0

Sapsed, J., Grantham, A., & DeFillippi, R. (2007). A bridge over troubled waters: Bridging organisations and entrepreneurial opportunities in emerging sectors. *Research Policy, 36*(9), 1314–1334.

Schwartz, M., & Hornych, C. (2010). Cooperation patterns of incubator firms and the impact of incubator specialization: Empirical evidence from Germany. *Technovation, 30*(9), 485–495. https://doi.org/10.1016/j.technovation.2010.05.001

Shane, S., & Stuart, T. (2002). Organizational endowments and the performance of university start-ups. *Management Science, 48*(1), 154–170. https://doi.org/10.1287/mnsc.48.1.154.14280

Shepherd, D. A., Douglas, E. J., & Shanley, M. (2000). New venture survival: Ignorance, external shocks, and risk reduction strategies. *Journal of Business Venturing, 15*(5–6), 393–410. https://doi.org/10.1016/S0883-9026(98)00032-9

Siegel, D. S., Westhead, P., & Wright, M. (2003a). Assessing the impact of university science parks on research productivity: Exploratory firm-level evidence from the United Kingdom. *International Journal of Industrial Organization, 21*(9), 1357–1369. https://doi.org/10.1016/S0167-7187(03)00086-9

Siegel, D. S., Westhead, P., & Wright, M. (2003b). Science parks and the performance of new technology-based firms: A review of recent UK evidence and an agenda for future research. *Small Business Economics, 20*(2), 177–184.

Sivasubramaniam, N., Liebowitz, S., & Lackman, C. (2012). Determinants of new product development team performance: A meta-analytic review. *Journal of Product Innovation Management, 29*(5), 803–820. https://doi.org/10.1111/j.1540-5885.2012.00940.x

Sofouli, E., & Vonortas, N. S. (2007). S&T Parks and business incubators in middle-sized countries: The case of Greece. *The Journal of Technology Transfer, 32*(5), 525–544. https://doi.org/10.1007/s10961-005-6031-1

Squicciarini, M. (2008). Science Parks' tenants versus out-of-park firms: Who innovates more? A duration model. *The Journal of Technology Transfer, 33*(1), 45–71. https://doi.org/10.1007/s10961-007-9037-z

Squicciarini, M. (2009). Science parks: Seedbeds of innovation? A duration analysis of firms' patenting activity. *Small Business Economics, 32*(2), 169–190. https://doi.org/10.1007/s11187-007-9075-9a

Thompson, J. (1967). *Organizations in action*. McGraw-Hill.

Tingley, D., Yamamoto, T., Hirose, K., Keele, L., & Imai, K. (2014). Mediation: R package for causal mediation analysis. http://hdl.handle.net/https://doi.org/10.18637/jss.v059.i05

Tortoriello, M., & Krackhardt, D. (2010). Activating cross-boundary knowledge: The role of Simmelian ties in the generation of innovations. *Academy of Management Journal, 53*(1), 167–181. https://doi.org/10.5465/amj.2010.48037420

Ucbasaran, D., Lockett, A., Wright, M., & Westhead, P. (2003). Entrepreneurial founder teams: Factors associated with member entry and exit. *Entrepreneurship Theory and Practice, 28*(2), 107–128.

Van der Borgh, M., Cloodt, M., & Romme, A. G. L. (2012). Value creation by knowledge-based ecosystems: Evidence from a field study. *R&D Management, 42*(2), 150–169. https://doi.org/10.1111/j.1467-9310.2011.00673.x

Vanacker, T., Collewaert, V., & Zahra, S. A. (2017). Slack resources, firm performance, and the institutional context: Evidence from privately held European firms. *Strategic Management Journal, 38*(6), 1305–1326. https://doi.org/10.1002/smj.2583

Van Weele, M., van Rijnsoever, F. J., & Nauta, F. (2017). You can't always get what you want: How entrepreneur's perceived resource needs affect the incubator's assertiveness. *Technovation, 59*, 18–33. https://doi.org/10.1016/j.technovation.2016.08.004

Vásquez-Urriago, Á. R., Barge-Gil, A., & Rico, A. M. (2016a). Which firms benefit more from being located in a science and technology park? *Empirical Evidence for Spain. Research Evaluation, 25*(1), 107–117. https://doi.org/10.1093/reseval/rvv033

Vásquez-Urriago, Á. R., Barge-Gil, A., & Rico, A. M. (2016b). Science and technology parks and cooperation for innovation: Empirical evidence from Spain. *Research Policy, 45*(1), 137–147. https://doi.org/10.1016/j.respol.2015.07.006

Vásquez-Urriago, Á. R., Barge-Gil, A., Rico, A. M., & Paraskevopoulou, E. (2014). The impact of science and technology parks on firms' product innovation: Empirical evidence from Spain. *Journal of Evolutionary Economics, 24*(4), 835–873. https://doi.org/10.1007/s00191-013-0337-1

Westhead, P. (1997). RandD inputs and outputs of technology-based firms located on and off science parks. *R&D Management, 27*(1), 45–62. https://doi.org/10.1111/1467-9310.00041

Westhead, P. (2021). *Science parks*. Edward Elgar Publishing.

Westhead, P., & Batstone, S. (1998). Independent technology-based firms: The perceived benefits of a science park location. *Urban Studies, 35*(12), 2197–2219.

Westhead, P., & Batstone, S. (1999). Perceived benefits of a managed science park location. *Entrepreneurship & Regional Development, 11*(2), 129–154. https://doi.org/10.1080/089856299283236

Weston, R., & Gore, P. (2006). A brief guide to structural equation modeling. *The Counseling Psychologist, 34*(5), 719–751. https://doi.org/10.1177/0011000006286345

Wickham, H. (2011). ggplot2. *Wiley Interdisciplinary Reviews: Computational Statistics, 3*(2), 180–185.

Wiersema, M. F., & Bantel, K. A. (1992). Top management team demography and corporate strategic change. *Academy of Management Journal, 35*(1), 91–121. https://doi.org/10.5465/256474

Wiklund, J., Baker, T., & Shepherd, D. (2010). The age-effect of financial indicators as buffers against liability of newness. *Journal of Business Venturing, 25*(4), 423–437.

Williams, L. J., Hartman, N., & Cavazotte, F. (2010). Method variance and marker variables: A review and comprehensive CFA marker technique. *Organizational Research Methods, 13*(3), 477–514. https://doi.org/10.1177/1094428110366036

Woolley, J. L., & MacGregor, N. (2021). The influence of incubator and accelerator participation on nanotechnology venture success. *Entrepreneurship Theory and Practice*.

Yan, A., & Louis, M. R. (1999). The migration of organizational functions to the work unit level: Buffering, spanning, and bringing up boundaries. *Human Relations, 52*(1), 25–47.

Yli-Renko, H., Autio, E., & Sapienza, H. J. (2001). Social capital, knowledge acquisition, and knowledge exploitation in young technology-based firms. *Strategic Management Journal, 22*(6–7), 587–613. https://doi.org/10.1002/smj.183

Zeng, S. X., Xie, X. M., & Tam, C. M. (2010). Relationship between cooperation networks and innovation performance of SMEs. *Technovation, 30*(3), 181–194. https://doi.org/10.1016/j.technovation.2009.08.003

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